

Online Retail Sales Data Engineering

Dataset: Obtain [a sample online retail sales dataset with information](#) about products, customers, and sales transactions.

Tasks:

Data Ingestion:

- Implement a data ingestion process to fetch the dataset from its source (e.g., CSV file, database).
- Load the dataset into a suitable storage system (e.g., relational database, data warehouse).

Data Cleaning and Validation:

- Develop a data cleaning pipeline to handle missing values, duplicates, and outliers.
- Validate the integrity and consistency of the data during the cleaning process.

Data Transformation:

- Transform the data into a structured format suitable for analytical processing.
- Create a pipeline to convert relevant columns to appropriate data types.
- Implement any necessary data standardization or normalization.

Data Warehouse Design:

- Design and create a data warehouse schema to support efficient querying and reporting.
- Define tables for products, customers, and sales transactions.
- Establish appropriate relationships between tables.

ETL (Extract, Transform, Load) Process:

- Build an ETL pipeline to automate the extraction, transformation, and loading of data.
- Schedule the ETL process to run at regular intervals (e.g., daily, hourly).

Partitioning and Indexing:

- Implement partitioning and indexing strategies to optimize data retrieval performance.
- Evaluate and choose suitable columns for partitioning and indexing.

Data Quality Monitoring:

- Set up monitoring mechanisms to track data quality over time.
- Establish alerts for anomalies or issues detected during the ETL process.

Data Versioning:

- Implement a versioning system for the dataset to track changes and updates.
- Archive previous versions of the dataset for historical analysis.

Metadata Management:

- Create a metadata repository to document the structure and lineage of the dataset.
- Include information on data sources, transformations, and storage locations.

Performance Optimization:

- Optimize the performance of data retrieval queries by analysing query execution plans.
- Implement caching mechanisms for frequently accessed data.

Deliverables:

- Python or SQL scripts for each step of the ETL process.
- Documentation outlining the data warehouse schema, ETL pipeline, and data quality monitoring.
- Recommendations for further optimizations and improvements.